Structural analysis – Level 3 – Forced response analysis – VS pump

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1. PURPOSE & SCOPE

This procedure is related to the dynamic analysis requirements for structural pumps, especially to the forced response analysis.

This procedure describes how to perform a structural response frequency (modal) analysis on vertically suspended (VS) pumps. It can also include flexible elements, such as piping supports or other flexible mounting elements. This may be necessary when suitable separation margins provided in ~~of~~ **SOP-482** – *"Pump structural natural frequency analysis - VS pump"* cannot be met through reasonable structural modifications.

The procedure for determining the required level of detail in dynamic analysis, in alignment with good practices for pump design (*ANSI/HI 9.6.8-2014)* is explained in **G.10.00.09**, and the parameters are outlined in **Mod.401**. According to the procedure, if calculated RUN is higher than 50 and less than or equal to 160, a **Level 3 – Advanced method – Structural analysis – Forced response analysis - VS pump** is required.

This procedure is applicable to VS pumps. It can also be extended to OH and BB pumps vertically mounted.

This procedure is applicable to Nova Milanese plant and La Spezia plant.

2. DEFINITIONS

B, b Damping

BB Between Bearings (API 610)

BDF Book Data File (solver Input file)

e Eccentricity

e_{per} Permissible eccentricity

FEA Finite Element Analysis

G Balance quality grade

K, k Stiffness

M, m Mass

N,n Rotating speed

OH Overhung (API 610)

P, p Load

R Risk number (ANSI HI 9.6.8-2014)

RUN R multiplied by U (ANSI HI 9.6.8-2014)

t Time

U Uncertainty value (ANSI HI 9.6.8-2014)

U Unbalance load (API610 & ISO 1940-1)

U_{per} Permissible unbalance (API610 & ISO 1940-1)

VS Vertically suspended (API 610)

X, x Position

Ω Rotating speed

ω Excitation frequency

3. GENERAL PROCEDURE – DESCRIPTION

Frequency response analysis is a method used to compute the structural response of the steady-state oscillatory excitation.

Two different numerical methods can be used in frequency response analysis:

- The direct method solves the coupled equations of motion in terms of forcing frequency (not discussed).
- The modal method utilizes the mode shapes of the structure to reduce and uncouple the equations of motion; the solution for a particular forcing frequency is obtained through the summation of the individual modal responses.

In **direct frequency response analysis**, a structural response is computed at discrete excitation frequencies by solving a set of coupled matrix equations. The analysis starts with the damped forced vibration equation of motion with harmonic excitation:

$$[M]\{\ddot{x}(t)\} + [B]\{\dot{x}(t)\} + [K]\{x(t)\} = \{P(\omega)\}e^{i\omega t}$$

For harmonic motion, it is assumed to be a harmonic solution of the form:

$$\{x\} = \{u(\omega)\}e^{i\omega t}$$

Where $\{u(\omega)\}$ is a complex displacement vector.

Substituting $\{x\}$, and applying the first and second derivatives in the expression above, the following is obtained:

$$-\omega^2[M]\{u(\omega)\}e^{i\omega t} + i\omega[B]\{u(\omega)\}e^{i\omega t} + [K]\{u(\omega)\}e^{i\omega t} = \{P(\omega)\}e^{i\omega t}$$

Then, after dividing by $e^{i\omega t}$, the resulting simplified expression is obtained:

$$\left[-\omega^2 M + i\omega B + K\right]\{u(\omega)\} = \{P(\omega)\}$$

The equation of motion is solved by inserting the forcing frequency ω into the equation of motion. This expression represents a system of equations with complex coefficients if damping is included or the applied loads have phase angles. The equations of motions at each input frequency are then solved using complex arithmetic.

Modal frequency response analysis is an alternative approach to compute the frequency response of a structure. This method uses the mode shapes of the structure to reduce the size, uncouple the equations of motions (when modal or no damping is used), and make the numerical model more efficient. Since the mode shapes are typically computed as part of the characterization of the structure, modal frequency response analysis is a natural extension of a normal modes analysis.

As a first step, the variables are transformed from physical coordinates $\{u(\omega)\}$ to modal coordinates $\{\xi\}$ by assuming:

$$\{x\} = [\phi]\{\xi\}e^{i\omega t}$$

Where $[\phi]$ represent the mode shapes.

The analysis continues temporarily ignoring the damping, which results in the undamped equation for harmonic motion at forcing frequency ω.

$$-\omega^2[M]\{x\} + [K]\{x\} = \{P(\omega)\}$$

Substituting the modal coordinates with the physical coordinates and dividing by $e^{i\omega t}$, the following is obtained:

$$-\omega^2[M][\phi]\{\xi\} + [K][\phi]\{\xi\} = \{P(\omega)\}$$

The equations remain coupled. To uncouple the equations, pre- multiply by $[\phi]^T$ to obtain:

$$-\omega^2[\phi]^T[M][\phi]\{\xi\} + [\phi]^T[K][\phi]\{\xi\} = [\phi]^T\{P(\omega)\}$$

Where:

- $[\phi]^T[M][\phi]$ is the modal (generalized) mass matrix.
- $[\phi]^T[K][\phi]$ is the modal (generalized) stiffness matrix.
- $[\phi]^T\{P\}$ is the modal force vector.

The final step uses the orthogonality property of the mode shapes to formulate the equation of motion in terms of the generalized mass and stiffness matrices, which are diagonal matrices. These diagonal matrices do not have the off-diagonal terms that couple the equations of motion. Therefore, the modal equations of motion are uncoupled and can be written as a set of uncoupled single degree-of-freedom system as:

$$-\omega^2 m_i \xi_i(\omega) + k_i \xi_i(\omega) = p_i(\omega)$$

Where:

- m_i is the ith modal mass.
- k_i is the ith modal stiffness.
- p_i is the ith modal force.

The modal form of the frequency response analysis reduces to a series of uncoupled single degree-of-freedom systems.

Once the individual modal responses $\xi_i(\omega)$ are computed, physical responses are recovered as the summation of the modal responses using:

$$\{x\} = [\phi]\{\xi\}e^{i\omega t}$$

If a damping matrix $[B]$ exists, the orthogonality property of the modes does not, in general, diagonalize the general damping matrix.

$$[\phi]^T[B][\phi] \neq diagonal$$

If structural damping is used, the orthogonality property does not, in general, diagonalize the generalized stiffness matrix.

$$[\phi]^T[K][\phi] \neq diagonal$$

Where $K = (1 + iG)[K] + i\Sigma G_E[K_E]$.

In the presence of a $[B]$ matrix or a complex stiffness matrix, the modal frequency approach solves the coupled problem in terms of modal coordinates using the direct frequency approach.

$$\left[-\omega^2[\phi]^T[M][\phi] + i\omega[\phi]^T[B][\phi] + [\phi]^T[K][\phi]\right]\{\xi(\omega)\} = [\phi]^T\{P(\omega)\}$$

If damping is applied to each mode separately, the uncoupled equation of motion can be maintained. When modal damping is used, each mode has damping b_i where $b_i = 2m_i\omega_i\zeta_i$.

The equations of motion remain uncoupled and have the following form for each mode:

$$-\omega^2 m_i\xi_i(\omega) + i\omega b_i\xi_i(\omega) + k_i\xi_i(\omega) = p_i(\omega)$$

Each of the modal responses is computed using:

$$\xi_i(\omega) = \frac{p_i(\omega)}{-\omega^2 m_i + i\omega b_i + k_i}$$

It is possible that not all the computed modes are required in the frequency response solution. The minimum number of modes to retain is equal to the number whose resonant frequencies lie withing the range of forcing frequencies. This guideline is only a minimum requirement. For better accuracy, all modes up to at least two to three times the highest forcing frequency should be retained.

4. METHODOLOGY, ANALYSIS AND INTERPRETATION OF RESULTS

Methodology

Software/Tools:

- General purpose FEA programs.

In this procedure, MSC Software (ANNEX A. MODAL FREQUENCY RESPONSE ANALYSIS (MSC NASTRAN)) will be used to carry out the structural dynamic analysis.

The tools described related to MSC Software are:

- MSC Nastran.

Model

See **SOP-482** – *"Level 2 Pump structural natural frequency analysis - VS pump"* for model information.

Analysis should include damping (primarily from the pump impeller oscillating and dissipating energy based on experience). A modal damping of 1% is a typical value.

Analysis

Forced response analysis should evaluate displacements and stresses of the system subjected to a mechanical unbalance load according to Chapter *0 Permissible residual unbalance* or other load test.

In the analysis, forces should be applied using the worst-case scenario of location and direction.

Structural modes and corresponding natural frequencies are calculated with an undamped modal analysis. The structure should be investigated including the liquid mass.

Forced response analysis should include modes, displacement, and stress results.

Interpretation of results

Among the significant modes, there are bending modes of the aboveground structure ("Reed" modes of discharge head, driver stand and motor) and the columns. In some cases, the components above and below the floor are dynamically coupled. In all cases, 1x synchronous speed excitation due to mechanical and hydraulic excitation is an excitation source. Bearing whirl excitation at approximately 50% running speed may also be considered as an excitation source, particularly for the column modes.

Because of the low structural damping typically present with VS pumps, even low power applications are subject to excessive structural vibrations in case of resonance. Displacements and stresses should be evaluated at critical locations along the structure as a mean of assessment.

Permissible residual unbalance

The permissible residual unbalance U_{per}, according to ISO 1940-1, can be derived based on a selected balance quality grade G according to the following equation:

$$U_{per} = 1000 \cdot \frac{(e_{per} \cdot \Omega) \cdot m}{\Omega} = 9549.3 \cdot \frac{G \cdot m}{\mathrm{N}}$$

Where:

- U_{per} is the numerical value of the permissible residual unbalance, [g*mm].
- e_{per}*Ω (or G) is the numerical value of the selected balance quality grade, [mm/s].
- m is the numerical value of the component mass, [kg].
- Ω (or N) is the numerical value of the angular velocity of the service speed, [rad/s or rpm].

API formula is approximately equal to the midpoint of the corresponding ISO balance quality grade, and it is very restrictive (*Table 1*).

Table 1. Residual unbalance.

G	ISO 1940	API 610
0.67	$U_{per} = 9549 \times G \times \frac{m}{N}$	$U_{per} = 6350 \times \frac{m}{N}$
1.0	$U_{per} = 9549 \times G \times \frac{m}{N}$	$U_{per} = 1.5 \times 6350 \times \frac{m}{N}$
2.5	$U_{per} = 9549 \times G \times \frac{m}{N}$	$U_{per} = 4 \times 6350 \times \frac{m}{N}$

5. ACCEPTANCE CRITERIA

ANSI HI 9.6.4 refers to the evaluation of vibration when the vibration measurements are made on stationary parts associated with bearings. It provides the maximum allowable vibration values in field and factory environments.

Vibration measurements shall be located at the motor mounting flanges.

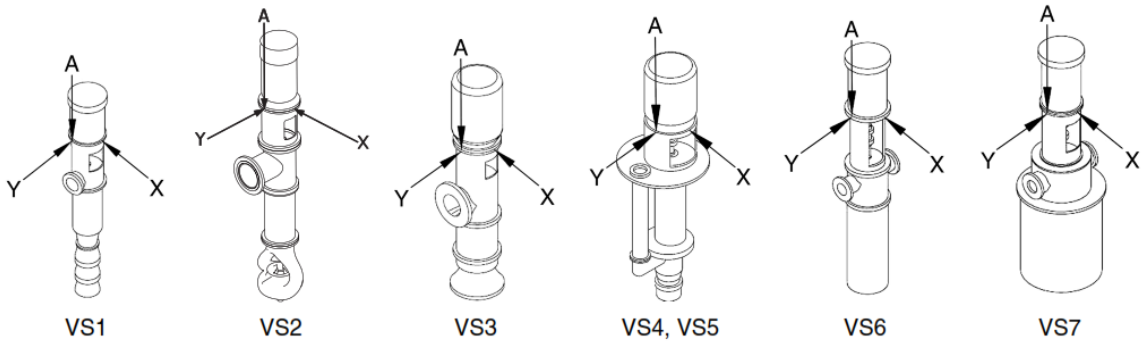


Figure 1. Measurement locations and directions.

The allowable pump bearing housing vibration is reported in *Figure 2*.

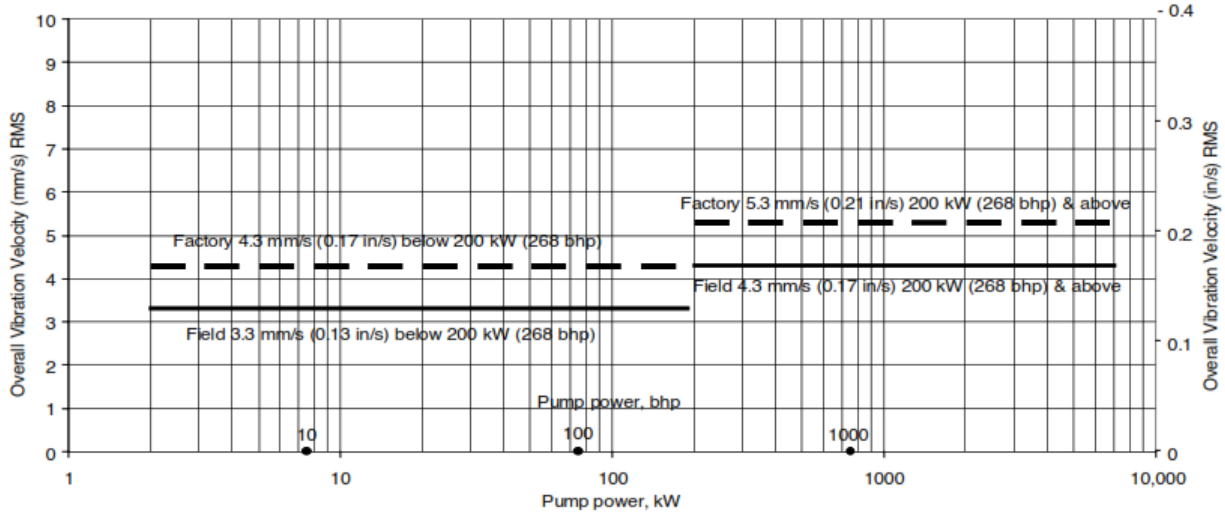


Figure 2. Allowable pump vibration – VS pumps.

For speeds of 600 rpm and below, both the allowable velocity vibration criteria reported in the figure above and the allowable displacement vibration criteria (*Table 2*) shall be met.

Table 2. Additional acceptance criteria for pumps operating at 600 rpm and below.

Operating Condition & Application Data	Field peak-to-peak at running speed	Factory peak-to-peak at running speed
Preferred operating region	80 µm	100 µm
Allowable operating region	100 µm	125 µm
An additional value will only be applied when the measurement location is greater than 1.5 m above the foundation. In this case, the additional value is determined per meter of height that exceeds 1.5 m above the foundation. Example: Measurement location = 2 m above the foundation Additive value = 50 µm/m x 0.5 m = 25 µm	50 µm	

Table 3 highlights the acceptance criteria for the forced response analysis according to ANSI HI 9.6.4 standard. Numerical results at forcing frequency should be lower than the values in *Table 3*.

Table 3. Acceptance criteria - VS Pumps.

	$P < 200\ kW$	$P > 200\ kW$
Overall velocity	$v_u < 3.3\ mm/s\ RMS$	$v_u < 4.3\ mm/s\ RMS$
Overall displacement (Speed ≤ 600 rpm)	$A_u < 80\ \mu m\ peak-to-peak$ If measurement location exceeds the height 1.5 m above the foundation: $A_u < 80 + (50\ [\mu m/m] \cdot H[m])\ \mu m\ peak-to-peak$	
Where: • v_u is the overall velocity resulting from numerical analysis. • A_u is the amplitude of overall displacement resulting from numerical analysis. • H is the height that exceeds 1.5 m above the foundation.		

6. REFERENCES

API STANDARD 610.

ANSI/HI 9.6.8-2014 Rotordynamic Pumps – Guideline for Dynamic Pumping Machinery.

ANSI/HI 9.6.4-2016 Rotordynamic Pumps for Vibration Measurements and Allowable Values.

MSC Nastran 2024.1, Windows documentation.

G.10.00.09 - Guidelines for dynamic analysis complexity level.

Mod.401 – Spreadsheet linked to procedure G.10.00.09 (in agreement with ANSI HI 9.6.8)

Mod.542 – FRA Graph Results.

7. ANNEX A. MODAL FREQUENCY RESPONSE ANALYSIS (MSC NASTRAN)

Since frequency response analysis is usually performed after evaluating mode shapes that characterize the structure, a modal frequency response analysis can be prepared by modifying the BDF of the normal modes analysis.

A.1. *Structure MSC Nastran File*

To perform an analysis using MSC Nastran, an input file describing the structure's geometry, material properties, boundary conditions, and loads is required. In addition to defining the physical structure, the input file also specifies the type of analysis to be performed and other pertinent information. The input file is a text file (ASCII).

An MSC Nastran input file (*Figure 4*) consists of five distinct sections:

- **The NASTRAN statement**. It is an optional statement used to specify system parameters that control internal solution processing or provide specific diagnostics. If used, it must be placed before the Executive Control Statement. It is generally not required, but it is useful when specification of system-related parameters is needed.
- **The File Management Section (FMS).** If needed, it is located before the Executive Control Section. The purpose is to assign files, manipulate the database, and perform restarts.
- **The Executive Control Section** contains statements that:
  - select the solution sequence to be run (e.g., statics, normal modes).
  - set the maximum MSC Nastran (not system) time permitted for the run.
  - request various diagnostic outputs.

  The end of the Executive Control Section is denoted by the CEND statement.
- **The Case Control Section.** The commands are used to:
  - define subcases (e.g., loading and boundary conditions).
  - make selections from the Bulk Data Section (e.g., loads, constraints, etc.).
  - specify output requests.
  - define titles, subtitles and labels for documenting the analysis.
- **The Bulk Data Section** is used to define the analysis via the grid points, elements, constraints, and loads. The start of the Bulk Data Section is denoted by the BEGIN BULK

delimiter. The last entry in Bulk Data Section must be an ENDDATA delimiter. The ENDDATA delimiter also signifies the end of the MSC Nastran input file. If any of these required delimiters are not present, a fatal error message appears when the job is submitted.

Most of the entries in the Bulk Data Section do not need to be selected by a Case Control command to be included in the model.

Input in the File Management Section and the Executive Control Section is defined as a statement; input in the Case Control Section is defined as a command; and input in the Bulk Data Section is defined as an entry.

A.2. Executive Control Section, SOL number

In the Executive Control Section, change the SOL number that specifies the solution sequence to be executed. Normal modes analysis has SOL 103 while, modal frequency response analysis has SOL 111.

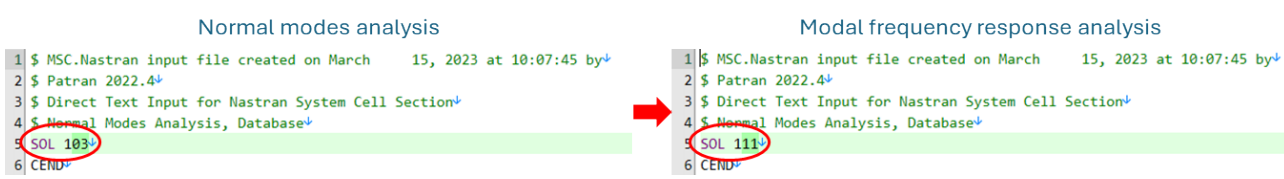


Figure 3. SOL input number.

```
$     FILENAME - TRUSS1.DAT
$
ID LINEAR,TRUSS1
SOL 101
TIME 2
CEND
```

Executive Control Section

```
TITLE = LINEAR STATICS USER'S SAMPLE INPUT FILE
SUBTITLE = TRUSS STRUCTURE
LABEL = POINT LOAD AT GRID POINT 4
LOAD = 10
SPC = 11
DISPLACEMENT = ALL
ELFORCE = ALL
ELSTRESS = ALL
```

Case Control Section

```
BEGIN BULK
$
$ THE GRID POINTS LOCATIONS
$ DESCRIBE THE GEOMETRY
$
GRID    1               0.      0.      0.              3456
GRID    2               0.      120.    0.              3456
GRID    3               600.    120.    0.              3456
GRID    4               600.    0.      0.              3456
$
$ MEMBERS ARE MODELED USING
$ ROD ELEMENTS
$
CROD    1       21      2       3
CROD    2       21      2       4
CROD    3       21      1       3
CROD    4       21      1       4
CROD    5       21      3       4
$
$ PROPERTIES  OF ROD ELEMENTS
$
PROD    21      22      4.      1.27
$
$ MATERIAL PROPERTIES
$
MAT1    22      30.E6           .3
$
$ POINT LOAD
$
FORCE   10      4               1000.   0.      -1.     0.
$
SPC1    11      123456  1       2
$
ENDDATA
```

Bulk Data Section

Figure 4. Example Input file.

A.3. Case Control Section, FREQUENCY, SDAMPING and DLOAD commands

In the Case Control Section, add the commands:

- $FREQUENCY = 1$
- $SDAMPING = 1$
- $DLOAD = 2$

FREQUENCY select the set of forcing frequencies to be solved in the frequency response problem. It calls FREQ, FREQ1, FREQ4, … in the Bulk Data Section.

SDAMPING request modal damping as a function of natural frequency. It calls TABDMP1 entry in the Bulk Data Section.

DLOAD select the dynamic load to apply in the analysis. Set identification of a DLOAD, or RLOAD2 in the Bulk Data section.

Normal modes analysis

```
CEND
ECHO = NONE
$ Using Nastran default values for RESVEC
MPC = 398821
   BCONTACT = 0
   BCONCHK(PLOT)=RUN
   $**********************
   AUTOSPC(NOZERO) = YES
   $MFLUID = 1
   $PARAM VMOPT 2
   PARAM,AUTOMSET,YES
   MEFFMASS(ALL) = YES
   $**********************
SUBCASE 1
   SUBTITLE=modal_rev000
   METHOD = 1
   BCONTACT = 1
   SPC = 2
   VECTOR(PLOT,SORT1,REAL)=ALL
   SPCFORCES(PLOT,SORT1,REAL)=ALL
   ESE(PLOT)=ALL
$ Direct Text Input for this Subcase
BEGIN BULK
```

Modal frequency response analysis

```
CEND
ECHO = NONE
$ Using Nastran default values for RESVEC
MPC = 398821
   BCONTACT = 0
   BCONCHK(PLOT)=RUN
   $**********************
   AUTOSPC(NOZERO) = YES
   $MFLUID = 1
   $PARAM VMOPT 2
   PARAM,AUTOMSET,YES
   MEFFMASS(ALL) = YES
   $**********************
SUBCASE 1
   SUBTITLE=rfa_1
   METHOD = 1
   FREQUENCY = 1
   SDAMPING = 1
   DLOAD = 2
   BCONTACT = 1
   SPC = 2
   DISPLACEMENT(PLOT,SORT1,REAL)=ALL
   VELOCITY (PLOT,SORT1,REAL)=ALL
   ACCELERATION (PLOT,SORT1,REAL)=ALL
   SPCFORCES(PLOT,SORT1,REAL)=ALL
   STRESS(PLOT,SORT1,REAL,VONMISES,BILIN)=ALL
   MPCFORCES(PLOT)=ALL
   OLOAD(PLOT)=ALL
$ Direct Text Input for this Subcase
BEGIN BULK
```

Figure 5. FREQUENCY, SDAMPING and DLOAD commands.

A.4. Case Control Section, Output results

In the Case Control Section, add the following commands for the output requests:

- $DISPLACEMENT(PLOT,SORT1,REAL) = ALL$
- $VELOCITY(PLOT,SORT1,REAL) = ALL$
- $ACCELERATION(PLOT,SORT1,REAL) = ALL$
- $STRESS(PLOT,SORT1,REAL,VONMISES,BILIN) = ALL$
- $MPCFORCES(PLOT) = ALL$
- $OLOAD(PLOT) = ALL$

Delete the commands:

- $VECTOR(PLOT,SORT1,REAL) = ALL$
- $ESE(PLOT) = ALL$

$DISPLACEMENT$, $VELOCITY$, $ACCELERATION$ request the form and type of displacement, velocity and acceleration vector output. $VECTOR$ request the form and type of displacement vector output, similar as $DISPLACEMENT$.

$STRESS$ requests the form and type of element stress output.

$MPCFORCES$, $SPCFORCES$ request respectively the form and type of multipoint force of constraint and single point force of constraint vector output.

$OLOAD$ requests the form and type of applied load vector output.

ESE requests the output of the strain energy in selected elements.

Table 4. Optional parameters.

Describer	Meaning
$PLOT$	Generate the database. Specify in the Bulk Data Section the entry “HDF5OUT” to create a h5 binary database file.
$SORT1$	Output will be presented as a tabular listing of grid points for each load, or frequency.
$REAL$	Requests rectangular format (magnitude and phase) of complex output.
$VONMISES$	Requests von Mises stresses.
$BILIN$	Request surface elements stress at center and grid points using bilinear extrapolation.

ALL	Output for all grid points/elements.

Normal modes analysis

```
CEND
ECHO = NONE
$ Using Nastran default values for RESVEC
MPC = 398821
   BCONTACT = 0
   BCONCHK(PLOT)=RUN
   $**********************
   AUTOSPC(NOZERO) = YES
   $MFLUID = 1
   $PARAM VMOPT 2
   PARAM,AUTOMSET,YES
   MEFFMASS(ALL) = YES
   $**********************
SUBCASE 1
   SUBTITLE=modal_rev000
   METHOD = 1
   BCONTACT = 1
   SPC = 2
   VECTOR(PLOT,SORT1,REAL)=ALL
   SPCFORCES(PLOT,SORT1,REAL)=ALL
   ESE(PLOT)=ALL
$ Direct Text Input for this Subcase
BEGIN BULK
```

Modal frequency response analysis

```
CEND
ECHO = NONE
$ Using Nastran default values for RESVEC
MPC = 398821
   BCONTACT = 0
   BCONCHK(PLOT)=RUN
   $**********************
   AUTOSPC(NOZERO) = YES
   $MFLUID = 1
   $PARAM VMOPT 2
   PARAM,AUTOMSET,YES
   MEFFMASS(ALL) = YES
   $**********************
SUBCASE 1
   SUBTITLE=rfa_1
   METHOD = 1
   FREQUENCY = 1
   SDAMPING = 1
   DLOAD = 2
   BCONTACT = 1
   SPC = 2
   DISPLACEMENT(PLOT,SORT1,REAL)=ALL
   VELOCITY (PLOT,SORT1,REAL)=ALL
   ACCELERATION (PLOT,SORT1,REAL)=ALL
   SPCFORCES(PLOT,SORT1,REAL)=ALL
   STRESS(PLOT,SORT1,REAL,VONMISES,BILIN)=ALL
   MPCFORCES(PLOT)=ALL
   OLOAD(PLOT)=ALL
$ Direct Text Input for this Subcase
BEGIN BULK
```

Figure 6. Output result commands.

The h5 output file size depends on the number of grid points and elements stored in it. For big numerical models (i.e. greater than 500000 elements) the computational time required for the calculation can increase considerably. In these cases, it is suggested to create in the Case Control Section two SET commands, one for the nodes and one for the elements, as showed in *Figure 7*.

SET is used to define lists of ID for grid points or elements for processing and output request. Each set has its identification number and the list of nodes/elements. The nodes/elements can be a convenient sub-system of the entire model.

In the commands of output results, associate the SET defined with them. To $DISPLACEMENT$, $VELOCITY$, $ACCELERATION$, $MPCFORCES$ and $LOAD$, associate the set containing nodes, while to $STRESS$, associate the set containing elements.

```
CEND
ECHO = NONE
$1------2-------3-------4-------5-------6-------7-------8-------9-------
$ Elements (and connectors) for group : B01)RFA_rev001
SET 99 = 1 THRU 53768,54489 THRU 57241,70749 THRU 70948,92029,92030,
92032,92033,92034,94555,94556,124236 THRU 124543,131057 THRU 135881,
136616 THRU 149425,151479 THRU 151566,151567
$ Nodes for group : B01)RFA_rev001
SET 88 = 29 THRU 38,122 THRU 310,374 THRU 394,2202 THRU 2210,2214,2653,
2654 THRU 2850,2873 THRU 2894,4522 THRU 4710,4774 THRU 4794,6622,
6623 THRU 6810,6874 THRU 6894,10340 THRU 10349,10435 THRU 10623,10687,
10688 THRU 10707,12578 THRU 12586,12590,12680 THRU 12877,13208,
13209 THRU 13229,14967 THRU 15155,15219 THRU 15239,17130 THRU 17318,
17382 THRU 17402,20408 THRU 20609,45162,45163,45169,45221 THRU 45230,
45318,45330,45377 THRU 45385,45472 THRU 45483,45597 THRU 45725,46460,
46461 THRU 46468,46694 THRU 46813,47494 THRU 47613,48204 THRU 48311,
49152 THRU 49312,49359 THRU 49381,49405 THRU 49427,49474 THRU 50452,
50465,50466,50536,50537,50595 THRU 50608,50621,50690 THRU 50704,51506,
51507 THRU 52213,52774 THRU 54306,54482 THRU 54490,54516 THRU 54524,
54549 THRU 54558,54611 THRU 54620,54647 THRU 54664,54683 THRU 56035,
61060 THRU 61909,61932 THRU 61940,61947,61990 THRU 61999,62021,
62022 THRU 62026,62032,62033,62034,62035,62057 THRU 62066,62088,
62089 THRU 62787,63474 THRU 64829,69368 THRU 69653,69674 THRU 69702,
70300 THRU 76144,76813,76814,76815,76816,78071 THRU 84704,85261,85262,
85263,85264,89861,89979,90183,90344,93307 THRU 93370,93502 THRU 98043,
98386 THRU 100085,100646 THRU 100711,100716 THRU 100760,100765,
100766 THRU 100815,100820 THRU 100864,100869 THRU 103001,103405,
103406 THRU 103525,103718 THRU 103932,104124 THRU 104271,104463,
104464 THRU 112797,112846,112847,112848,112849,112854 THRU 112892,
112898 THRU 112910,112915 THRU 112919,112924 THRU 112944,112950,
112951 THRU 112975,112980 THRU 113031,113036 THRU 113083,115624,
115625 THRU 116024,122059 THRU 125981,126026 THRU 126221,126884,
126885 THRU 140488,141036 THRU 141165,142465 THRU 142670,142896,
142897 THRU 142986,144164 THRU 163151,163152
$ Using Nastran default values for RESVEC
```

Figure 7. Nastran SET.

```
$ Output Requests
DISPLACEMENT(PLOT,SORT1,REAL)=88
VELOCITY (PLOT,SORT1,REAL)=88
ACCELERATION (PLOT,SORT1,REAL)=88
STRESS(PLOT,SORT1,REAL,VONMISES,BILIN)=99
```

Figure 8.Output results commands with SET.

A.5. Bulk Data Section, Real Eigenvalue Analysis

Check the frequency range defined (EIGRL) has a sufficient number of modes. For good accuracy, all modes up to at least two to three times the highest forcing frequency should be retained.

Normal modes analysis

```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$*************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$*************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$*************************
EIGRL,1,0.,100.
```

Modal frequency response analysis

```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$*************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$*************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$*************************
EIGRL,1,0.,100.
$ Frequency step
$1------2-------3-------4-------5-------6-------7-------8-------9-------
FREQ     1       12.3
FREQ1    1       5.     1.      20
FREQ4    1       5.     25.     .01     7
```

$EIGRL$ defines data needed to perform real eigenvalue analysis (vibration).

Table 5. EIGRL Entry Format.

1	2	3	4	5	6	7	8	9
EIGRL	SID	V1	V2					

Table 6. EIGRL Input Entry.

Describer	Meaning
SID	Set identification number. It must be selected with the Case Control command METHOD = SID (Integer > 0).
V1, V2	Frequency range of interest in cycles per unit time. Set V1 equal to 0.0 and V2 equal to 2 or 3 times the highest forcing frequency (Real ≥ 0.0).

A.6. Bulk Data Section, Forcing Frequencies

In the Bulk Data Section, add the following entries with the appropriate inputs:

- $FREQ$
- $FREQ1$
- $FREQ4$

Normal modes analysis

```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$************************
EIGRL,1,0.,100.
BCPARA,0,METHOD,SEGTOSEG
$************************
BCTABL1  0       1       10      11      12      13      14      15
         16      17      2       3       39      4       5       6
         7       8       9       37      38      25      26      27
         28      30      31      32      33      34      35      36
         29
```

Modal frequency response analysis

```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$************************
EIGRL,1,0.,100.
$ Frequency step
$1------2-------3-------4-------5-------6-------7-------8-------9-------
FREQ     1       12.3
FREQ1    1       5.      1.      20
FREQ4    1       5.      25.     .01     7
$*****************************************************************
$ Modal mass damping
$1------2-------3-------4-------5-------6-------7-------8-------9-------
TABDMP1  1       CRIT
         0.      .01     1000.   .01     ENDT
$1------2-------3-------4-------5-------6-------7-------8-------9-------
DLOAD    2       1.      1.      302     1.      303
$*****************************************************************
TABLED1  1
         0.      0.      100.    100.    ENDT
$ Impeller unbalance
$1------2-------3-------4-------5-------6-------7-------8-------9-------
RLOAD2   302     402     0.      0.      1       0.      0
RLOAD2   303     403     0.      90.     1       0.      0
DAREA    402     99049   1       31.
DAREA    403     99049   2       31.
$*****************************************************************
BCPARA,0,METHOD,SEGTOSEG
$************************
BCTABL1  0       1       10      11      12      13      14      15
         16      17      2       3       39      4       5       6
         7       8       9       37      38      25      26      27
         28      30      31      32      33      34      35      36
         29
```

Figure 9. Set of frequencies entries.

$FREQ$ defines a set of frequencies to be used in the solution of frequency response problems.

Table 7. FREQ Entry Format.

1	2	3	4	5	6	7	8	9
FREQ	SID	F1	F2	F3	F4	F5	F6	F7
	F8	F9	F10	-etc.-				

Table 8. FREQ Input Entry.

Describer	Meaning
SID	Set identification number. It must be selected with the Case Control command FREQUENCY = SID (Integer > 0).
Fi	Frequency value in units of cycles per unit time. Include at least the rotating frequency to study (Real ≥ 0.0).

$FREQ1$ defines a set of frequencies to be used in the solution of frequency response problems by specifying a starting frequency, frequency increment, and the number of increments desired.

Define a range around the rotating frequency to evaluate.

Table 9. FREQ1 Entry Format.

1	2	3	4	5	6	7	8	9
FREQ	SID	F1	DF	NDF				

Table 10. FREQ1 Input Entry.

Describer	Meaning
SID	Set identification number. It must be selected with the Case Control command FREQUENCY = SID (Integer > 0).
F1	First frequency in set in units of cycles per unit time (Real ≥ 0.0).
DF	Frequency increment in set in units of cycles per unit time (Real ≥ 0.0).
NDF	Number of frequency increments (Integer > 0).

The frequencies defined by FREQ1 entry are given by:

$$f_i = F1 + DF \cdot (i - 1)$$

Where $i = 1$ to $NDF + 1$.

$FREQ4$ defines a set of frequencies used in the solution of modal frequency-response problems by specifying the amount of "spread" around each natural frequency and the number of equally spaced excitation frequencies within the spread.

Define a range between F1 and F2 around the rotating frequency to evaluate.

Table 11. FREQ4 Entry Format.

1	2	3	4	5	6	7	8	9
FREQ4	SID	F1	F2	FSPD	NFM			

Table 12. FREQ4 Input Entry.

Describer	Meaning
SID	Set identification number. It must be selected with the Case Control command FREQUENCY = SID (Integer > 0).
F1	Lower bound of frequency range in units of cycles per unit time (Real ≥ 0.0).
F2	Upper bound of frequency range in cycles per unit time (Real ≥ 0.0; F2 > F1).
FSPD	Frequency spread. There will be NFM excitation frequencies between $(1 - FSPD) \cdot f_N$ and $(1 + FSPD) \cdot f_N$ for each natural frequency in the range F1 to F2. (1.0 > Real > 0.0). Set FSPD equal to the damping ratio (0.1).
NFM	Number of evenly spaced frequencies per "spread" mode. Set NFM at least equal to 5, 7 or 9 as a good compromise between calculation time and accuracy (Integer > 0).

The frequency spread can be used also to define the half-power bandwidth. The half-power bandwidth is given by $2 \cdot \xi \cdot f_N$, where ξ is the damping ratio. If FSPD is specified equal to the damping ratio for the mode, NFM specifies the number of excitation frequencies within the half-power bandwidth.

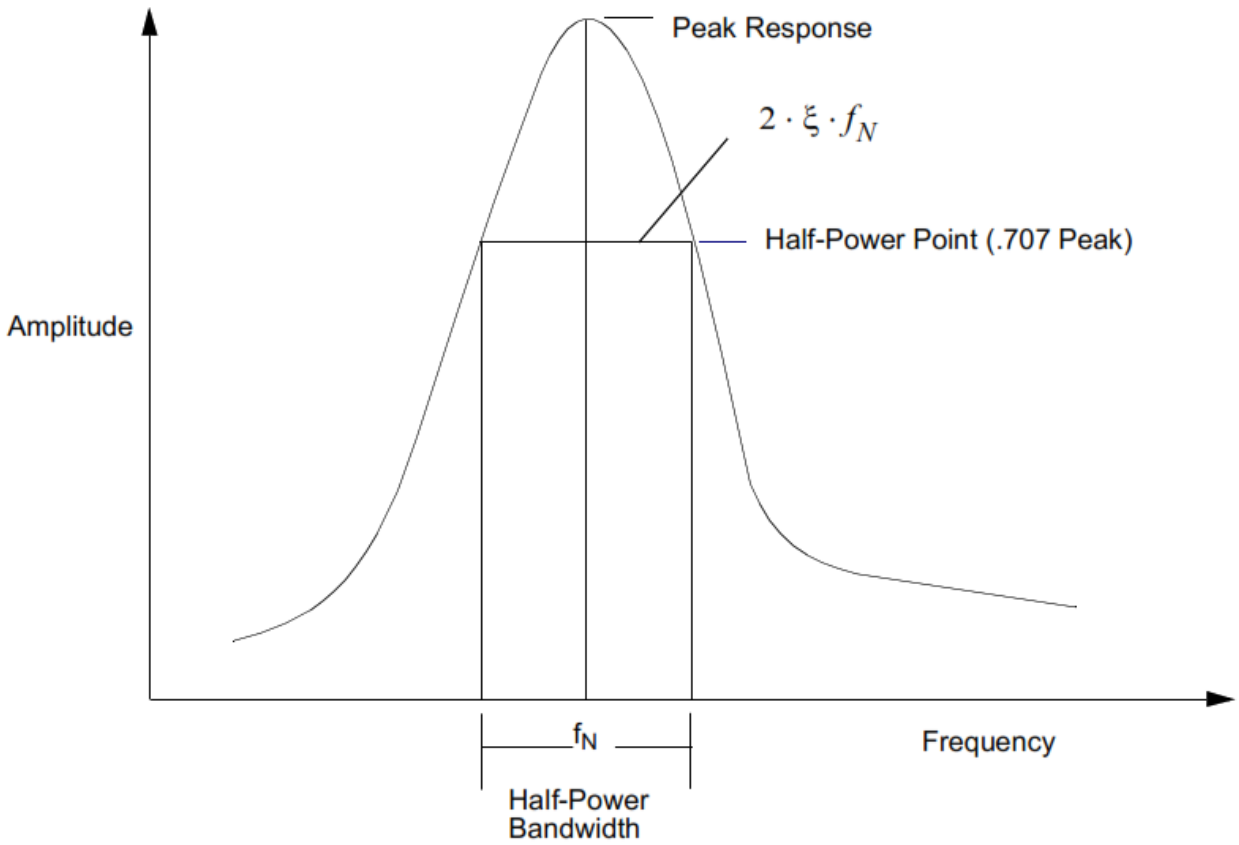


Figure 10. Half-Power Bandwidth Definition.

A.7. Bulk Data Section, Modal Damping

In the Bulk Data Section, add the $TABDMP1$ entry following the format in *Table 13*. It defines modal damping as a tabular function of natural frequency.

To be precautionary, define a critical damping of 1% in all the frequency range (i.e. 0 to 1000 Hz).

Normal modes analysis

```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$************************
EIGRL,1,0.,100.
BCPARA,0,METHOD,SEGTOSEG
$************************
BCTABL1  0       1       10      11      12      13      14      15
         16      17      2       3       39      4       5       6
         7       8       9       37      38      25      26      27
         28      30      31      32      33      34      35      36
         29
```

Modal frequency response analysis

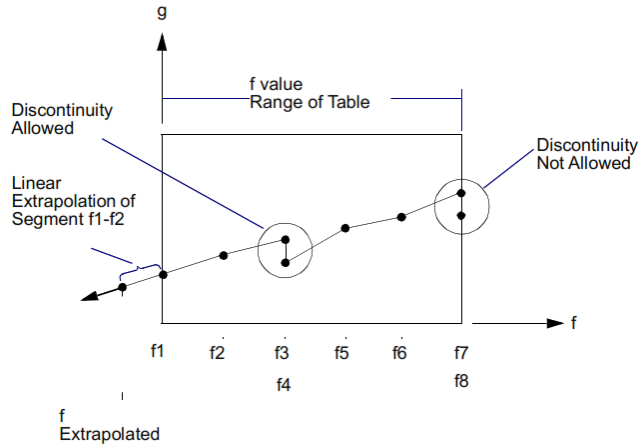
```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$************************
EIGRL,1,0.,100.
$ Frequency step
$1------2-------3-------4-------5-------6-------7-------8-------9-------
FREQ     1       12.3
FREQ1    1       5.      1.      20
FREQ4    1       5.      25.     .01     7
$*********************************************************************
$ Modal mass damping
$1------2-------3-------4-------5-------6-------7-------8-------9-------
TABDMP1  1       CRIT
         0.      .01     1000.   .01     ENDT
$1------2-------3-------4-------5-------6-------7-------8-------9-------
DLOAD    2       1.      1.      302     1.      303
$*********************************************************************
TABLED1  1
         0.      0.      100.    100.    ENDT
$ Impeller unbalance
$1------2-------3-------4-------5-------6-------7-------8-------9-------
RLOAD2   302     402     0.      0.      1       0.      0
RLOAD2   303     403     0.      90.     1       0.      0
DAREA    402     99049   1       31.
DAREA    403     99049   2       31.
$*********************************************************************
BCPARA,0,METHOD,SEGTOSEG
$************************
BCTABL1  0       1       10      11      12      13      14      15
         16      17      2       3       39      4       5       6
         7       8       9       37      38      25      26      27
         28      30      31      32      33      34      35      36
         29
```

Figure 11. Modal damping table.

Table 13. TABDMP1 Entry Format.

1	2	3	4	5	6	7	8	9
TABDMP1	TID	TYPE						
	f1	g1	f2	g2	f3	g3	-etc.-	ENDT

Table 14. TABDMP1 Input Entry.

Describer	Meaning
TID	Table identification number. It must be selected with the Case Control command SDAMPING = TID (Integer > 0).
TYPE	Type of damping units: • G units of equivalent viscous dampers. • CRIT units of fraction of critical damping (suggested). • Q units of the amplification factor.
fi	Natural frequency values in cycles per unit time. They must be specified in either ascending or descending order, but not both. Discontinuities may be specified between any two points except the two starting points (Real ≥ 0.0).  *Figure 12. Example of Table extrapolation and discontinuity.*
gi	Damping value. Suggested value is 0.1 of critical damping (Real ≥ 0.0).
ENDT	Flag indicating the end of the table.

A.8. Bulk Data Section, Dynamic Load

In the Bulk Data Section, add the entries to define an applied load (force or moment):

- *DLOAD*
  - *RLOAD2*
    - *DAREA*
    - *TABLED1*

Normal modes analysis

```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$*************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$*************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$*************************
EIGRL,1,0.,100.
BCPARA,0,METHOD,SEGTOSEG
$*************************
BCTABL1  0       1       10      11      12      13      14      15
         16      17      2       3       39      4       5       6
         7       8       9       37      38      25      26      27
         28      30      31      32      33      34      35      36
         29
```

Modal frequency response analysis

```
BEGIN BULK
HDF5OUT PRCISION 32     CMPRMTHD LZ4     LEVEL   5
$*************************
include 'CFAST.bdf'
$include 'MFLUID.bdf'
$*************************
MDLPRM,IGNSHBDN,1
PARAM   PRTMAXIM YES
$*************************
EIGRL,1,0.,100.
$ Frequency step
$1------2-------3-------4-------5-------6-------7-------8-------9-------
FREQ     1       12.3
FREQ1    1       5.      1.      20
FREQ4    1       5.      25.     .01     7
$*********************************************************************
$ Modal mass damping
$1------2-------3-------4-------5-------6-------7-------8-------9-------
TABDMP1  1       CRIT
         0.      .01     1000.   .01     ENDT
$1------2-------3-------4-------5-------6-------7-------8-------9-------
DLOAD    2       1.      1.      302     1.      303
$*********************************************************************
TABLED1  1
         0.      0.      100.    100.    ENDT
$ Impeller unbalance
$1------2-------3-------4-------5-------6-------7-------8-------9-------
RLOAD2   302     402     0.      0.      1       0.      0
RLOAD2   303     403     0.      90.     1       0.      0
DAREA    402     99049   1       31.
DAREA    403     99049   2       31.
$*********************************************************************
BCPARA,0,METHOD,SEGTOSEG
$*************************
BCTABL1  0       1       10      11      12      13      14      15
         16      17      2       3       39      4       5       6
         7       8       9       37      38      25      26      27
         28      30      31      32      33      34      35      36
         29
```

Figure 13. Dynamic load entries.

DLOAD defines a dynamic loading condition as a linear combination of load sets defined via RLAOD2.

Table 15. DLAOD Entry Format.

1	2	3	4	5	6	7	8	9
DLAOD	SID	S	S1	L1	S2	L2	S3	L3
	S4	L4	-etc.-					

Table 16. DLAOD Input Entry.

Describer	Meaning
SID	Load set identification number. It must be selected with the Case Control command DLAOD = SID (Integer > 0).
S	Scale factor. Use as default 1.0 (Real).
Si	Scale factors. Use as default 1.0 (Real).
Li	Load set identification numbers of RLOAD2 entries. Each Li must be unique from any other Li on the same entry (Integer > 0).

The load vector being defined by this entry is given by:

$$\{P\} = S \cdot \sum_i Si \cdot \{P_i\}$$

Add scale factors Si and load set identification number Li based on the loads to apply.

$RLOAD2$ defines a frequency dependent dynamic excitation of the form represented in the equation below. It is used in frequency response problems.

$$\{P(f)\} = \{A\} \cdot B(f) e^{i\{\phi(f) + \theta - 2\pi f \tau\}}$$

Table 17. RLAOD2 Entry Format.

1	2	3	4	5	6	7	8	9
RLAOD2	SID	EXCITEDID	DELAYR	DPHASER	TB	RP	TYPE	

Table 18. RLAOD2 Input Entry.

<table>
<tr><th>Describer</th><th>Meaning</th></tr>
<tr><td>SID</td><td>Set identification number. It must be unique for all dynamic load entries. The DLOAD Li entry number will select all dynamic load entries with set identification number SID equal to Li. (Integer > 0).</td></tr>
<tr><td>EXCITED</td><td>Entry set that defines $\{A\}$. (Integer > 0).
See TYPE for entry.</td></tr>
<tr><td>DELAYR</td><td>Value of time delay τ. In general, set time delay equal to 0.0 (Real).</td></tr>
<tr><td>DPHASER</td><td>Value of phase angle θ. In general, set phase angle equal to 0.0 or 90.0 (Real).</td></tr>
<tr><td>TB</td><td>Set identification number of TABLEDi entry that gives $B(f)$ (Integer > 0).</td></tr>
<tr><td>RP</td><td>Value of ϕ to be used for all frequencies. In general, set ϕ equal to 0.0. (Real).</td></tr>
<tr><td>TYPE</td><td>Defines the type of the dynamic excitation. (Integer, character).
<table>
<tr><th>TYPE</th><th>Type of Dynamic excitation</th></tr>
<tr><td>0, L, LO, LOA or LOAD</td><td>Applied load (force or moment)</td></tr>
<tr><td>1, D, DI, DIS or DISP</td><td>Enforced displacement using SPC data</td></tr>
<tr><td>2, V, VE, VEL or VELO</td><td>Enforced velocity using SPC data</td></tr>
<tr><td>3, A, AC, ACC, ACCE</td><td>Enforced acceleration using SPC data</td></tr>
</table>
It also determines the manner in which EXCITED is used. If the specified dynamic excitation type is 0 (applied load), then:
• EXCITED may reference to DAREA.
If the specified dynamic excitation type is 1, 2 or 3 (enforced motion), then:
• EXCITED may reference to SPCD.</td></tr>
</table>

$DAREA$ defines the scale (area) factors for static and dynamic loads. In dynamic analysis, it is used in conjunction with RLOADi entries.

Table 19. DAREA Entry Format.

1	2	3	4	5	6	7	8	9
DAREA	SID	P1	C1	A1	P2	C2	A2	

Table 20. DAREA Input Entry.

Describer	Meaning
SID	Identification number. The RLAOD2 EXCITED entry number will select the identification number SID of DAREA (Integer > 0).
Pi	Grid point identification number (Integer > 0).
Ci	Component number of displacement coordinate system (load direction) (Integer 1 through 6).
Ai	Scale (area) factor $\{A\}$ (Real).

$TABLED1$ defines a tabular function to use in generating frequency dependent dynamic loads.

Use it to define $B(f)$ of RLOAD2 as a function of frequency.

Table 21. TABLED1 Entry Format.

1	2	3	4	5	6	7	8	9
TABLED1	TID	XAXIS	YAXIS					
	x1	y1	x2	y2	x3	y3	-etc.-	ENDT

Table 22. TABLED1 Input Entry.

Describer	Meaning
TID	Table identification number (Integer > 0).
XAXIS	Specifies a linear or logarithmic interpolation for the x-axis (Character: LINEAR or LOG, if blank linear interpolation).
YAXIS	Specifies a linear or logarithmic interpolation for the y-axis (Character: LINEAR or LOG, if blank linear interpolation).
xi, yi	Tabular values (Real). In this case x values will be the frequency and y values will be $B(f)$ of RLOAD2.
ENDT	Flag indicating the end of the table.

The most common and easiest load to define in forced response analysis is mechanical unbalance. It is proportional to the rotating speed (frequency) and it is suggested for precaution to assume a value of 4 times the permissible unbalance load:

$$\{P\} = 4 \cdot U_{per} \cdot (2\pi f)^2$$

Where U_{per} is defined in 0

Permissible residual *unbalance* as:

$$U_{per} = 1000 \cdot \frac{G \cdot m}{(2\pi f)}$$

Resulting in:

$$\{P\} = 4 \cdot 1000 \cdot G \cdot m \cdot (2\pi f)$$

Since RLOAD2 defines an excitation as:

$$\{P(f)\} = \{A\} \cdot B(f) e^{i\{\theta\}}$$

It allows to define an unbalance load as:

- constant at different frequencies.
- proportional to rotating frequency.

For constant unbalance results:

- $\{A\} = 4 \cdot 1000 \cdot G \cdot m \cdot (2\pi f)$ where f is the rated frequency.
- $B(f) = 1$ → $TABLED1$ is a constant function at different frequencies.

For an unbalance proportional to the rotating speed results:

- $\{A\} = 4 \cdot 1000 \cdot G \cdot m \cdot (2\pi)$
- $B(f) = f$ → $TABLED1$ is a ramp function proportional to the frequency.

ATTENTION! RLOAD2 defines a sinusoidal load in one direction specified in the $DAREA$. To generate a rotating load, two RLOAD2 must be defined with two different directions. Moreover, the loads should have a phase difference (DPHASER) of 90°. See *Figure 13* for better understanding.

8. ANNEX B. MOD.542 – FRA GRAPH RESULTS

Mod.542 – *"FRA Graph Results"* is a Python script. It processes and visualizes data from an HDF5 file. It requires different parameters as input (*Figure 14*):

- The h5 filename that store analysis results, called **filename.**
- A flag for the magnitude of motion result, called **ordinate**. Input one between:
  - *RMS*
  - *0-pk*
  - *pk-pk*
- A flag for plot displacement results, called **displ**. Input between *True* to plot displacements or *False* to not plot.
- A flag for plot velocity results, called **vel**. Input between *True* to plot velocity or *False* to not plot it.
- A flag for plot velocity results, called **acc**. Input between *True* to plot acceleration or *False* to not plot it.
- The list of nodes for which create the graphs, called **nodelist**. It is important to add node IDs between square brackets and each one separated by a comma.

```
#--INPUT DATA-----------------------------------------------------------------------------------
filename = "t30719001_rev000_rfa.h5"                # name of h5 file
ordinate = "RMS"                                    # Flag for RMS, 0-pk or pk-pk magnitude
displ = False                                       # Flag for displacement (True or False)
vel = True                                          # Flag for velocity (True or False)
acc = False                                         # Flag for acceleration (True or False)
nodelist = [99049, 99045]                           # List of nodes which extract graphs
#-----------------------------------------------------------------------------------------------
```

Figure 14. Mod.542 - Graph Results Input Parameters.

After running the script, it will give as output an image and an output file for each node.

Each image (*Figure 15*) contains the amplitude of the motion in the upper plot, while the phase angle in the frequency range studied for the three directions of the numerical model coordinate system is located in the lower plot.

The maximum value in the amplitude plot at the separation margin of the rotating speed should not exceed the value reported in Chapter *5 ACCEPTANCE CRITERIA*

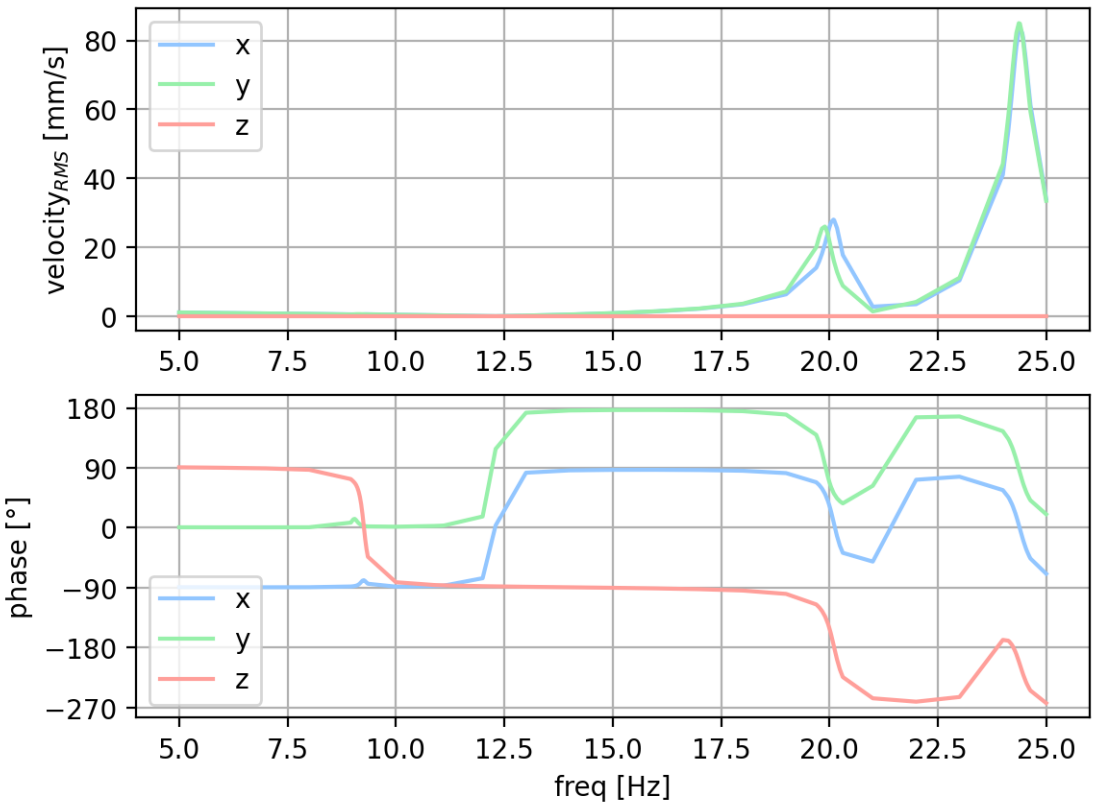


Figure 15. Amplitude and phase magnitude plot.

The output is a text file that contains the same data as the image. It can be used to create customized graphics if required.